■ Lab 15 — Advanced Terraform & CI/CD (AWS CloudShell)

■ Objectifs du Lab

- Utiliser le **dependency lock file**, le bloc `moved`, le bloc `check`
- Gérer les **secrets** de manière sécurisée
- Utiliser `terraform plan -out=planfile` + `terraform apply planfile`
- Mettre en place un pipeline **CI/CD GitHub Actions**

■ Étape 1 — Préparer l'environnement

```
bash $HOME/install-terraform.sh
mkdir -p $HOME/terraform-training/lab15-advanced
cd $HOME/terraform-training/lab15-advanced
```

> *Le reste des instructions Terraform est identique à la version principale (README.md).*

■ Étape 2 — Créer le .gitignore AVANT tout

> ■ ***Critique — à faire avant tout `git add`.***

```
cat > .gitignore << 'EOF'
.terraform/
tfplan
tfplan-destroy
*.tfplan
terraform.tfstate
terraform.tfstate.backup
*.auto.tfvars
EOF
```

■ Étape 3 — Cloner le repo GitHub

```
git config --global user.email "<votre-email>"
git config --global user.name "<votre-prenom>"

git clone https://<votre-token>@github.com/<votre-compte>/tf-training-<votre-prenom>.git
cd tf-training-<votre-prenom>
```

■ Étape 4 — Créer le workflow à la racine du repo

■■ *Le dossier `.github/workflows/` doit être à la **racine du repo**.*

```
# Se placer à la racine du repo cloné
cd $HOME/terraform-training/tf-training-<votre-prenom>

mkdir -p .github/workflows

cat > .github/workflows/terraform.yml << 'EOF'
name: Terraform CI/CD

on:
  push:
    branches: [ main ]
    paths: [ 'labs/lab15-advanced/**' ]
  pull_request:
    branches: [ main ]
    paths: [ 'labs/lab15-advanced/**' ]
  workflow_dispatch:
    inputs:
      action:
        description: 'Action à exécuter'
        required: true
        default: 'plan'
        type: choice
        options: [ plan, apply, destroy ]

env:
  TF_WORKING_DIR: labs/lab15-advanced
  TF_VAR_username: ${{ github.actor }}

jobs:
  terraform:
```

```
    name: Terraform ${{ github.event.inputs.action || 'plan' }}
    runs-on: ubuntu-latest
    defaults:
      run:
        working-directory: ${{ env.TF_WORKING_DIR }}
    steps:
      - name: Checkout code
        uses: actions/checkout@v4
      - name: Setup Terraform
        uses: hashicorp/setup-terraform@v3
        with:
          terraform_version: "1.15.5"
      - name: Configure AWS credentials
        uses: aws-actions/configure-aws-credentials@v4
        with:
          aws-access-key-id: ${{ secrets.AWS_ACCESS_KEY_ID }}
          aws-secret-access-key: ${{ secrets.AWS_SECRET_ACCESS_KEY }}
          aws-region: eu-west-1
      - name: Terraform Init
        run: terraform init
      - name: Terraform Format Check
        run: terraform fmt -check -recursive
      - name: Terraform Validate
        run: terraform validate
      - name: Terraform Plan
        run: terraform plan -var="environment=dev" -out=tfplan
        if: github.event_name == 'pull_request' || github.event.inputs.action == 'plan' || github.event_
      - name: Upload Plan
        uses: actions/upload-artifact@v4
        with:
          name: tfplan
          path: ${{ env.TF_WORKING_DIR }}/tfplan
          retention-days: 1
        if: github.event_name == 'pull_request' || github.event.inputs.action == 'plan' || github.event_
      - name: Terraform Apply
        run: terraform apply tfplan
        if: github.event.inputs.action == 'apply'
      - name: Terraform Destroy
        run: |
          terraform plan -destroy -var="environment=dev" -out=tfplan-destroy
          terraform apply tfplan-destroy
        if: github.event.inputs.action == 'destroy'
EOF
```

■ Étape 5 — Vérifier et commiter

```
# Vérifier ce qui sera commité
git status

# ■ Attendu : .github/, labs/lab15-advanced/*.tf, .terraform.lock.hcl
# ■ NE PAS voir : .terraform/, tfplan

git add .
git commit -m "feat: lab15 advanced terraform + CI/CD"
git push origin main
```

■ Validation du Lab

#	Critère	Validé
1	`.gitignore` créé avant le premier `git add`	■
2	`.terraform/` et `tfplan` absents du repo	■
3	`.github/workflows/` à la racine du repo	■
4	Pipeline GitHub Actions se déclenche sur push	■
5	Apply et Destroy via `workflow_dispatch` fonctionnent	■

■ **Fin du Lab 15 !**